

# Lunar Surface Infrastructure Program WBS Dictionary

*Level 3 scope definitions and control attributes version 0.1*

|                 |                     |               |                                   |
|-----------------|---------------------|---------------|-----------------------------------|
| <b>Owner</b>    | Program Director    | <b>Status</b> | Draft for review                  |
| <b>Version</b>  | 0.1                 | <b>Basis</b>  | Program Charter v1.1 and WBS v0.2 |
| <b>Coverage</b> | 40 Level 3 elements | <b>Date</b>   | 21 September 2026                 |

## Purpose

This dictionary converts the approved WBS structure into controlled scope definitions for the 40 Level 3 deliverables. It clarifies what each element contains, what it excludes, how completion will be judged, and what management controls must be established before execution.

## Control objective

Use this draft to review scope coverage, eliminate overlap, confirm accountable functions, identify dependencies and risks, and decide where Level 4 work-package decomposition is required. This document does not authorize funding, contractual commitments, crewed operations, or execution.

## Recommended decision

**Approve** the dictionary as the working scope-definition baseline after branch owners validate the entries and resolve comments. Then authorize selective Level 4 decomposition and integrated-master-schedule development.

# Dictionary Conventions

*How to interpret and control each entry*

## Entry fields

| Field               | Control use                                                                                         |
|---------------------|-----------------------------------------------------------------------------------------------------|
| Scope definition    | The outcome owned by the WBS element.                                                               |
| Included work       | The principal products, capabilities, analyses, and evidence contained in the element.              |
| Excluded work       | Work controlled elsewhere or intentionally outside the program baseline.                            |
| Acceptance criteria | Preliminary objective evidence required to declare the element complete.                            |
| Dependencies        | Inputs or predecessor capabilities required from other elements or providers.                       |
| Assumptions         | Planning conditions used until validated or replaced by controlled requirements.                    |
| Owner               | The preliminary accountable function; a named individual will be assigned during baseline planning. |
| Risk                | The dominant threat that could prevent or delay the deliverable outcome.                            |
| Decision gate       | The program review expected to accept the completion evidence.                                      |
| Decomposition       | Recommendation to retain the element as a control element or create Level 4 work packages.          |

## Global planning assumptions

- The program remains an educational planning model and does not represent an approved flight program.
- The lunar operating site is selected through controlled site-characterization evidence; a final site is not yet baselined.
- Robotic preparation precedes crew arrival, and crew survival and safe departure take priority over computing continuity.
- Common infrastructure is allocated under element 1.6 only when it supports more than one project.
- Dates, budgets, named owners, suppliers, launch services, and contractual authorities remain to be established.

# Level 3 Dictionary Index

*Forty controlled scope entries organized by Level 2 parent*

## **1.1 Program Integration and Systems Engineering**

- 1.1.1 Program Governance and Decision Management
- 1.1.2 Requirements and Configuration Baseline
- 1.1.3 Integrated Master Planning and Controls
- 1.1.4 Systems Architecture and Interface Management
- 1.1.5 Risk Opportunity and Issue Management
- 1.1.6 Verification Validation and Readiness Assurance
- 1.1.7 Stakeholder and Executive Communications

## **1.2 Launch and Lunar Transportation**

- 1.2.1 Mission Transportation Architecture
- 1.2.2 Payload Integration and Certification
- 1.2.3 Launch Services and Ground Operations
- 1.2.4 Trans Lunar and Lunar Orbit Operations
- 1.2.5 Descent Landing and Surface Delivery
- 1.2.6 Transportation Contingency and Recovery

## **1.3 Robotic Deployment and Site Preparation**

- 1.3.1 Robotic Fleet and Payload Configuration
- 1.3.2 Lunar Site Survey and Hazard Characterization
- 1.3.3 Surface Communications and Navigation Setup
- 1.3.4 Site Preparation and Grading
- 1.3.5 Infrastructure Deployment and Assembly
- 1.3.6 Inspection Maintenance and Remote Readiness Verification

## **1.4 Habitat and Crew Readiness**

- 1.4.1 Habitat Deployment and Commissioning
- 1.4.2 Environmental Control and Life Support
- 1.4.3 Crew Consumables and Waste Management
- 1.4.4 Radiation Dust and Thermal Protection
- 1.4.5 Crew Health Safety and Emergency Response
- 1.4.6 Crew Communications Operations and Safe Departure
- 1.4.7 Crew Mission Readiness and Training

## **1.5 Lunar Data Center**

- 1.5.1 Computing and Storage Architecture
- 1.5.2 Power Distribution and Energy Storage
- 1.5.3 Thermal Control and Heat Rejection
- 1.5.4 Communications and Data Transfer
- 1.5.5 Cybersecurity and Access Control
- 1.5.6 Autonomous Monitoring and Fault Management
- 1.5.7 Commissioning Performance Validation and Recovery

## **1.6 Shared Infrastructure and Resilience**

- 1.6.1 Common Surface Power
- 1.6.2 Shared Communications Navigation and Timing
- 1.6.3 Common Thermal Services
- 1.6.4 Logistics Spares and Maintenance Support
- 1.6.5 Dust Mitigation and Environmental Monitoring
- 1.6.6 Emergency Power and Contingency Operations
- 1.6.7 Integrated Resilience and Recovery Demonstration

# 1.1.1 Program Governance and Decision Management

Parent element 1.1 Program Integration and Systems Engineering

| WBS Dictionary Entry |                                                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the program governance and decision management capability required to support the program integration and systems engineering outcome.                                                                                                          |
| Included work        | Governance plan, review cadence, delegated authority, decision log, escalation path, and action closure.                                                                                                                                                  |
| Excluded work        | Project-specific design and execution remain within elements 1.2 through 1.6; this branch owns integration, governance, and program-level control. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Governance cadence, delegated authorities, and decision records are approved and operating.                                                                                                                                                               |
| Key dependencies     | Program Charter v1.1; technical and programmatic inputs from elements 1.2 through 1.6.                                                                                                                                                                    |
| Planning assumptions | Governance, requirements, interfaces, and evidence remain under formal configuration control.                                                                                                                                                             |
| Preliminary owner    | Program Director                                                                                                                                                                                                                                          |
| Primary risk         | Delayed decisions or inconsistent direction create baseline instability.                                                                                                                                                                                  |
| Decision gate        | Program Baseline Review                                                                                                                                                                                                                                   |
| Decomposition        | Candidate Level 3 level-of-effort control element; confirm measurable cadence and exit criteria in the baseline.                                                                                                                                          |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.1.2 Requirements and Configuration Baseline

Parent element 1.1 Program Integration and Systems Engineering

| WBS Dictionary Entry        |                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the requirements and configuration baseline capability required to support the program integration and systems engineering outcome.                                                                                                             |
| <b>Included work</b>        | Requirements hierarchy, traceability, configuration items, change control, and baseline status accounting.                                                                                                                                                |
| <b>Excluded work</b>        | Project-specific design and execution remain within elements 1.2 through 1.6; this branch owns integration, governance, and program-level control. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Program requirements and controlled configuration items are approved, traceable, and change-controlled.                                                                                                                                                   |
| <b>Key dependencies</b>     | Program Charter v1.1; technical and programmatic inputs from elements 1.2 through 1.6.                                                                                                                                                                    |
| <b>Planning assumptions</b> | Governance, requirements, interfaces, and evidence remain under formal configuration control.                                                                                                                                                             |
| <b>Preliminary owner</b>    | Requirements and Configuration Lead                                                                                                                                                                                                                       |
| <b>Primary risk</b>         | Requirement or interface changes propagate late rework across multiple projects.                                                                                                                                                                          |
| <b>Decision gate</b>        | Program Baseline Review                                                                                                                                                                                                                                   |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                                             |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.1.3 Integrated Master Planning and Controls

Parent element 1.1 Program Integration and Systems Engineering

| WBS Dictionary Entry        |                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the integrated master planning and controls capability required to support the program integration and systems engineering outcome.                                                                                                             |
| <b>Included work</b>        | Integrated scope, schedule architecture, resource plan, performance measures, forecasts, and baseline-control processes.                                                                                                                                  |
| <b>Excluded work</b>        | Project-specific design and execution remain within elements 1.2 through 1.6; this branch owns integration, governance, and program-level control. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Integrated scope, schedule, resources, and performance-measurement structure are baselined.                                                                                                                                                               |
| <b>Key dependencies</b>     | Program Charter v1.1; technical and programmatic inputs from elements 1.2 through 1.6.                                                                                                                                                                    |
| <b>Planning assumptions</b> | Governance, requirements, interfaces, and evidence remain under formal configuration control.                                                                                                                                                             |
| <b>Preliminary owner</b>    | Program Controls Manager                                                                                                                                                                                                                                  |
| <b>Primary risk</b>         | Incomplete logic, resource assumptions, or status data weakens forecast credibility.                                                                                                                                                                      |
| <b>Decision gate</b>        | Program Baseline Review                                                                                                                                                                                                                                   |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                                             |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.1.4 Systems Architecture and Interface Management

Parent element 1.1 Program Integration and Systems Engineering

| WBS Dictionary Entry        |                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the systems architecture and interface management capability required to support the program integration and systems engineering outcome.                                                                                                       |
| <b>Included work</b>        | System architecture, functional allocation, interface definitions, interface control documents, and technical trade decisions.                                                                                                                            |
| <b>Excluded work</b>        | Project-specific design and execution remain within elements 1.2 through 1.6; this branch owns integration, governance, and program-level control. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | System architecture and cross-project interfaces are defined, allocated, and accepted.                                                                                                                                                                    |
| <b>Key dependencies</b>     | Program Charter v1.1; technical and programmatic inputs from elements 1.2 through 1.6.                                                                                                                                                                    |
| <b>Planning assumptions</b> | Governance, requirements, interfaces, and evidence remain under formal configuration control.                                                                                                                                                             |
| <b>Preliminary owner</b>    | Chief Systems Engineer                                                                                                                                                                                                                                    |
| <b>Primary risk</b>         | Requirement or interface changes propagate late rework across multiple projects.                                                                                                                                                                          |
| <b>Decision gate</b>        | Program Baseline Review                                                                                                                                                                                                                                   |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                                             |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.1.5 Risk Opportunity and Issue Management

Parent element 1.1 Program Integration and Systems Engineering

| WBS Dictionary Entry |                                                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the risk opportunity and issue management capability required to support the program integration and systems engineering outcome.                                                                                                               |
| Included work        | Risk register, opportunity register, issue log, response plans, triggers, reserves, and escalation criteria.                                                                                                                                              |
| Excluded work        | Project-specific design and execution remain within elements 1.2 through 1.6; this branch owns integration, governance, and program-level control. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Program risks, opportunities, and issues have accountable owners, responses, and escalation paths.                                                                                                                                                        |
| Key dependencies     | Program Charter v1.1; technical and programmatic inputs from elements 1.2 through 1.6.                                                                                                                                                                    |
| Planning assumptions | Governance, requirements, interfaces, and evidence remain under formal configuration control.                                                                                                                                                             |
| Preliminary owner    | Program Risk Manager                                                                                                                                                                                                                                      |
| Primary risk         | Unrecognized triggers or optimistic response plans allow threats to mature into issues.                                                                                                                                                                   |
| Decision gate        | Program Baseline Review                                                                                                                                                                                                                                   |
| Decomposition        | Candidate Level 3 level-of-effort control element; confirm measurable cadence and exit criteria in the baseline.                                                                                                                                          |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.1.6 Verification Validation and Readiness Assurance

Parent element 1.1 Program Integration and Systems Engineering

| WBS Dictionary Entry |                                                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the verification validation and readiness assurance capability required to support the program integration and systems engineering outcome.                                                                                                     |
| Included work        | Verification matrix, validation approach, evidence plan, review entrance criteria, and readiness closure records.                                                                                                                                         |
| Excluded work        | Project-specific design and execution remain within elements 1.2 through 1.6; this branch owns integration, governance, and program-level control. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Verification strategy, evidence requirements, and decision-gate criteria are approved.                                                                                                                                                                    |
| Key dependencies     | Program Charter v1.1; technical and programmatic inputs from elements 1.2 through 1.6.                                                                                                                                                                    |
| Planning assumptions | Governance, requirements, interfaces, and evidence remain under formal configuration control.                                                                                                                                                             |
| Preliminary owner    | Verification and Readiness Lead                                                                                                                                                                                                                           |
| Primary risk         | Evidence gaps or ambiguous criteria delay authorization at a decision gate.                                                                                                                                                                               |
| Decision gate        | Program Baseline Review                                                                                                                                                                                                                                   |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                                             |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.1.7 Stakeholder and Executive Communications

Parent element 1.1 Program Integration and Systems Engineering

| WBS Dictionary Entry        |                                                                                                                                                                                                                                                           |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the stakeholder and executive communications capability required to support the program integration and systems engineering outcome.                                                                                                            |
| <b>Included work</b>        | Program reporting calendar, executive briefs, customer reviews, decision requests, and controlled status products.                                                                                                                                        |
| <b>Excluded work</b>        | Project-specific design and execution remain within elements 1.2 through 1.6; this branch owns integration, governance, and program-level control. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Program status, decisions, and forecasts are communicated through an approved reporting structure.                                                                                                                                                        |
| <b>Key dependencies</b>     | Program Charter v1.1; technical and programmatic inputs from elements 1.2 through 1.6.                                                                                                                                                                    |
| <b>Planning assumptions</b> | Governance, requirements, interfaces, and evidence remain under formal configuration control.                                                                                                                                                             |
| <b>Preliminary owner</b>    | Program Director                                                                                                                                                                                                                                          |
| <b>Primary risk</b>         | Delayed decisions or inconsistent direction create baseline instability.                                                                                                                                                                                  |
| <b>Decision gate</b>        | Program Baseline Review                                                                                                                                                                                                                                   |
| <b>Decomposition</b>        | Candidate Level 3 level-of-effort control element; confirm measurable cadence and exit criteria in the baseline.                                                                                                                                          |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.2.1 Mission Transportation Architecture

Parent element 1.2 Launch and Lunar Transportation

| WBS Dictionary Entry        |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the mission transportation architecture capability required to support the launch and lunar transportation outcome.                                                                                                      |
| <b>Included work</b>        | Mission phases, vehicle roles, payload flow, margins, trajectories, custody transfers, and transportation interfaces.                                                                                                              |
| <b>Excluded work</b>        | This branch ends after safe surface delivery and transfer of custody; development of a new launch vehicle remains excluded. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Transportation sequence, mission phases, vehicle roles, margins, and delivery interfaces are approved.                                                                                                                             |
| <b>Key dependencies</b>     | Approved payload manifest; launch-provider interfaces; mission architecture; landing and custody-transfer criteria.                                                                                                                |
| <b>Planning assumptions</b> | Launch and transportation services are procured; development of a new launch vehicle is excluded.                                                                                                                                  |
| <b>Preliminary owner</b>    | Mission Transportation Lead                                                                                                                                                                                                        |
| <b>Primary risk</b>         | Requirement or interface changes propagate late rework across multiple projects.                                                                                                                                                   |
| <b>Decision gate</b>        | Transportation Readiness Review                                                                                                                                                                                                    |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.2.2 Payload Integration and Certification

Parent element 1.2 Launch and Lunar Transportation

| WBS Dictionary Entry        |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the payload integration and certification capability required to support the launch and lunar transportation outcome.                                                                                                    |
| <b>Included work</b>        | Payload mechanical, electrical, software, environmental, safety, contamination, and certification evidence.                                                                                                                        |
| <b>Excluded work</b>        | This branch ends after safe surface delivery and transfer of custody; development of a new launch vehicle remains excluded. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Payloads satisfy mechanical, electrical, software, environmental, and safety integration requirements.                                                                                                                             |
| <b>Key dependencies</b>     | Approved payload manifest; launch-provider interfaces; mission architecture; landing and custody-transfer criteria.                                                                                                                |
| <b>Planning assumptions</b> | Launch and transportation services are procured; development of a new launch vehicle is excluded.                                                                                                                                  |
| <b>Preliminary owner</b>    | Payload Integration Lead                                                                                                                                                                                                           |
| <b>Primary risk</b>         | Mass, interface, trajectory, schedule, or landing margins may be insufficient for delivery.                                                                                                                                        |
| <b>Decision gate</b>        | Transportation Readiness Review                                                                                                                                                                                                    |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.2.3 Launch Services and Ground Operations

Parent element 1.2 Launch and Lunar Transportation

| WBS Dictionary Entry        |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the launch services and ground operations capability required to support the launch and lunar transportation outcome.                                                                                                    |
| <b>Included work</b>        | Launch contract interfaces, ground processing, range support, integrated procedures, rehearsals, and launch readiness.                                                                                                             |
| <b>Excluded work</b>        | This branch ends after safe surface delivery and transfer of custody; development of a new launch vehicle remains excluded. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Launch services, ground processing, range support, and launch-readiness evidence are accepted.                                                                                                                                     |
| <b>Key dependencies</b>     | Approved payload manifest; launch-provider interfaces; mission architecture; landing and custody-transfer criteria.                                                                                                                |
| <b>Planning assumptions</b> | Launch and transportation services are procured; development of a new launch vehicle is excluded.                                                                                                                                  |
| <b>Preliminary owner</b>    | Launch Operations Lead                                                                                                                                                                                                             |
| <b>Primary risk</b>         | Mass, interface, trajectory, schedule, or landing margins may be insufficient for delivery.                                                                                                                                        |
| <b>Decision gate</b>        | Transportation Readiness Review                                                                                                                                                                                                    |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.2.4 Trans Lunar and Lunar Orbit Operations

Parent element 1.2 Launch and Lunar Transportation

| WBS Dictionary Entry |                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the trans lunar and lunar orbit operations capability required to support the launch and lunar transportation outcome.                                                                                                   |
| Included work        | Transfer burns, guidance and navigation, communications coverage, orbit operations, tracking, and maneuver contingencies.                                                                                                          |
| Excluded work        | This branch ends after safe surface delivery and transfer of custody; development of a new launch vehicle remains excluded. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Transfer burns, navigation, communications, and lunar-orbit operations are planned and verified.                                                                                                                                   |
| Key dependencies     | Approved payload manifest; launch-provider interfaces; mission architecture; landing and custody-transfer criteria.                                                                                                                |
| Planning assumptions | Launch and transportation services are procured; development of a new launch vehicle is excluded.                                                                                                                                  |
| Preliminary owner    | Flight Operations Lead                                                                                                                                                                                                             |
| Primary risk         | Mass, interface, trajectory, schedule, or landing margins may be insufficient for delivery.                                                                                                                                        |
| Decision gate        | Transportation Readiness Review                                                                                                                                                                                                    |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.2.5 Descent Landing and Surface Delivery

Parent element 1.2 Launch and Lunar Transportation

| WBS Dictionary Entry        |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the descent landing and surface delivery capability required to support the launch and lunar transportation outcome.                                                                                                     |
| <b>Included work</b>        | Descent sequence, landing constraints, hazard avoidance, surface location, payload release, and custody-transfer evidence.                                                                                                         |
| <b>Excluded work</b>        | This branch ends after safe surface delivery and transfer of custody; development of a new launch vehicle remains excluded. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Assets are landed within approved constraints, located, and transferred to surface operations.                                                                                                                                     |
| <b>Key dependencies</b>     | Approved payload manifest; launch-provider interfaces; mission architecture; landing and custody-transfer criteria.                                                                                                                |
| <b>Planning assumptions</b> | Launch and transportation services are procured; development of a new launch vehicle is excluded.                                                                                                                                  |
| <b>Preliminary owner</b>    | Lunar Delivery Lead                                                                                                                                                                                                                |
| <b>Primary risk</b>         | Mass, interface, trajectory, schedule, or landing margins may be insufficient for delivery.                                                                                                                                        |
| <b>Decision gate</b>        | Transportation Readiness Review                                                                                                                                                                                                    |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.2.6 Transportation Contingency and Recovery

Parent element 1.2 Launch and Lunar Transportation

| WBS Dictionary Entry        |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the transportation contingency and recovery capability required to support the launch and lunar transportation outcome.                                                                                                  |
| <b>Included work</b>        | Delay, abort, missed-burn, off-nominal landing, asset safing, recovery, and alternate-delivery responses.                                                                                                                          |
| <b>Excluded work</b>        | This branch ends after safe surface delivery and transfer of custody; development of a new launch vehicle remains excluded. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Abort, delay, off-nominal delivery, and recovery responses are approved and exercised.                                                                                                                                             |
| <b>Key dependencies</b>     | Approved payload manifest; launch-provider interfaces; mission architecture; landing and custody-transfer criteria.                                                                                                                |
| <b>Planning assumptions</b> | Launch and transportation services are procured; development of a new launch vehicle is excluded.                                                                                                                                  |
| <b>Preliminary owner</b>    | Mission Assurance Lead                                                                                                                                                                                                             |
| <b>Primary risk</b>         | Mass, interface, trajectory, schedule, or landing margins may be insufficient for delivery.                                                                                                                                        |
| <b>Decision gate</b>        | Transportation Readiness Review                                                                                                                                                                                                    |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.3.1 Robotic Fleet and Payload Configuration

Parent element 1.3 Robotic Deployment and Site Preparation

| WBS Dictionary Entry |                                                                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the robotic fleet and payload configuration capability required to support the robotic deployment and site preparation outcome.                                                                                                  |
| Included work        | Robotic vehicles, mobility systems, tools, payloads, autonomy, spares, command modes, and operating concepts.                                                                                                                              |
| Excluded work        | This branch owns robotic surface execution and site preparation; shared utilities delivered for multiple projects remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Robotic vehicles, tools, payloads, spares, and operating concepts are configured and accepted.                                                                                                                                             |
| Key dependencies     | Verified surface delivery; site-selection criteria; robotic communications; utility and habitat interface requirements.                                                                                                                    |
| Planning assumptions | Robotic systems complete required preparation and verification before crew arrival.                                                                                                                                                        |
| Preliminary owner    | Robotics Project Manager                                                                                                                                                                                                                   |
| Primary risk         | Mass, interface, trajectory, schedule, or landing margins may be insufficient for delivery.                                                                                                                                                |
| Decision gate        | Robotic Site Readiness Review                                                                                                                                                                                                              |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                              |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.3.2 Lunar Site Survey and Hazard Characterization

Parent element 1.3 Robotic Deployment and Site Preparation

| WBS Dictionary Entry        |                                                                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the lunar site survey and hazard characterization capability required to support the robotic deployment and site preparation outcome.                                                                                            |
| <b>Included work</b>        | Terrain, slope, bearing strength, subsurface, illumination, thermal, dust, and obstacle characterization.                                                                                                                                  |
| <b>Excluded work</b>        | This branch owns robotic surface execution and site preparation; shared utilities delivered for multiple projects remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Terrain, subsurface, illumination, dust, and hazard data support an approved site decision.                                                                                                                                                |
| <b>Key dependencies</b>     | Verified surface delivery; site-selection criteria; robotic communications; utility and habitat interface requirements.                                                                                                                    |
| <b>Planning assumptions</b> | Robotic systems complete required preparation and verification before crew arrival.                                                                                                                                                        |
| <b>Preliminary owner</b>    | Site Characterization Lead                                                                                                                                                                                                                 |
| <b>Primary risk</b>         | Mobility limits, terrain uncertainty, dust, latency, or equipment faults may prevent site readiness.                                                                                                                                       |
| <b>Decision gate</b>        | Robotic Site Readiness Review                                                                                                                                                                                                              |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                              |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.3.3 Surface Communications and Navigation Setup

Parent element 1.3 Robotic Deployment and Site Preparation

| WBS Dictionary Entry        |                                                                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the surface communications and navigation setup capability required to support the robotic deployment and site preparation outcome.                                                                                              |
| <b>Included work</b>        | Command links, telemetry, positioning, navigation aids, time synchronization, coverage, and loss-of-link behavior.                                                                                                                         |
| <b>Excluded work</b>        | This branch owns robotic surface execution and site preparation; shared utilities delivered for multiple projects remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Robotic command, telemetry, positioning, and navigation services meet operating criteria.                                                                                                                                                  |
| <b>Key dependencies</b>     | Verified surface delivery; site-selection criteria; robotic communications; utility and habitat interface requirements.                                                                                                                    |
| <b>Planning assumptions</b> | Robotic systems complete required preparation and verification before crew arrival.                                                                                                                                                        |
| <b>Preliminary owner</b>    | Surface Network Lead                                                                                                                                                                                                                       |
| <b>Primary risk</b>         | Undefined scope, interfaces, or acceptance evidence may delay completion.                                                                                                                                                                  |
| <b>Decision gate</b>        | Robotic Site Readiness Review                                                                                                                                                                                                              |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                              |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.3.4 Site Preparation and Grading

Parent element 1.3 Robotic Deployment and Site Preparation

| WBS Dictionary Entry        |                                                                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the site preparation and grading capability required to support the robotic deployment and site preparation outcome.                                                                                                             |
| <b>Included work</b>        | Operating pads, travel routes, grading, excavation, berms, stabilization, clearance, and hazard removal.                                                                                                                                   |
| <b>Excluded work</b>        | This branch owns robotic surface execution and site preparation; shared utilities delivered for multiple projects remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Required operating areas and routes satisfy approved geometry, stability, and hazard limits.                                                                                                                                               |
| <b>Key dependencies</b>     | Verified surface delivery; site-selection criteria; robotic communications; utility and habitat interface requirements.                                                                                                                    |
| <b>Planning assumptions</b> | Robotic systems complete required preparation and verification before crew arrival.                                                                                                                                                        |
| <b>Preliminary owner</b>    | Site Preparation Lead                                                                                                                                                                                                                      |
| <b>Primary risk</b>         | Mobility limits, terrain uncertainty, dust, latency, or equipment faults may prevent site readiness.                                                                                                                                       |
| <b>Decision gate</b>        | Robotic Site Readiness Review                                                                                                                                                                                                              |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                              |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.3.5 Infrastructure Deployment and Assembly

Parent element 1.3 Robotic Deployment and Site Preparation

| WBS Dictionary Entry        |                                                                                                                                                                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the infrastructure deployment and assembly capability required to support the robotic deployment and site preparation outcome.                                                                                                   |
| <b>Included work</b>        | Equipment positioning, anchoring, assembly, cable or conduit routing, connection, alignment, and functional checks.                                                                                                                        |
| <b>Excluded work</b>        | This branch owns robotic surface execution and site preparation; shared utilities delivered for multiple projects remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Assigned equipment and infrastructure are positioned, connected, and functionally verified.                                                                                                                                                |
| <b>Key dependencies</b>     | Verified surface delivery; site-selection criteria; robotic communications; utility and habitat interface requirements.                                                                                                                    |
| <b>Planning assumptions</b> | Robotic systems complete required preparation and verification before crew arrival.                                                                                                                                                        |
| <b>Preliminary owner</b>    | Robotic Construction Lead                                                                                                                                                                                                                  |
| <b>Primary risk</b>         | Mobility limits, terrain uncertainty, dust, latency, or equipment faults may prevent site readiness.                                                                                                                                       |
| <b>Decision gate</b>        | Robotic Site Readiness Review                                                                                                                                                                                                              |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                              |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.3.6 Inspection Maintenance and Remote Readiness Verification

Parent element 1.3 Robotic Deployment and Site Preparation

| WBS Dictionary Entry |                                                                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the inspection maintenance and remote readiness verification capability required to support the robotic deployment and site preparation outcome.                                                                                 |
| Included work        | Inspection routes, condition assessment, corrective maintenance, exception closure, evidence packages, and remote approval.                                                                                                                |
| Excluded work        | This branch owns robotic surface execution and site preparation; shared utilities delivered for multiple projects remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | The site passes inspection, exceptions are dispositioned, and remote readiness is approved.                                                                                                                                                |
| Key dependencies     | Verified surface delivery; site-selection criteria; robotic communications; utility and habitat interface requirements.                                                                                                                    |
| Planning assumptions | Robotic systems complete required preparation and verification before crew arrival.                                                                                                                                                        |
| Preliminary owner    | Robotic Operations Lead                                                                                                                                                                                                                    |
| Primary risk         | Evidence gaps or ambiguous criteria delay authorization at a decision gate.                                                                                                                                                                |
| Decision gate        | Robotic Site Readiness Review                                                                                                                                                                                                              |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                              |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.4.1 Habitat Deployment and Commissioning

Parent element 1.4 Habitat and Crew Readiness

| WBS Dictionary Entry |                                                                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the habitat deployment and commissioning capability required to support the habitat and crew readiness outcome.                                                                                                       |
| Included work        | Habitat placement, anchoring, deployment, sealing, leak checks, activation, subsystem startup, and commissioning.                                                                                                               |
| Excluded work        | This branch owns crew-specific systems and readiness; common infrastructure serving multiple projects remains under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | The habitat is deployed, sealed, activated, and commissioned within approved limits.                                                                                                                                            |
| Key dependencies     | Verified site readiness; common utilities; transportation capability; approved crew concept and mission duration.                                                                                                               |
| Planning assumptions | The nominal mission supports seven crew members for 45 days with at least 60 days of critical provisions.                                                                                                                       |
| Preliminary owner    | Habitat Project Manager                                                                                                                                                                                                         |
| Primary risk         | A life-safety shortfall, contamination event, or insufficient margin may prevent crew authorization.                                                                                                                            |
| Decision gate        | Crewed Mission Readiness Review                                                                                                                                                                                                 |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                   |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.4.2 Environmental Control and Life Support

Parent element 1.4 Habitat and Crew Readiness

| WBS Dictionary Entry        |                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the environmental control and life support capability required to support the habitat and crew readiness outcome.                                                                                                     |
| <b>Included work</b>        | Atmosphere control, oxygen, carbon-dioxide removal, pressure, humidity, water processing, monitoring, and redundancy.                                                                                                           |
| <b>Excluded work</b>        | This branch owns crew-specific systems and readiness; common infrastructure serving multiple projects remains under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Atmosphere, pressure, oxygen, carbon-dioxide removal, humidity, and water systems meet crew criteria.                                                                                                                           |
| <b>Key dependencies</b>     | Verified site readiness; common utilities; transportation capability; approved crew concept and mission duration.                                                                                                               |
| <b>Planning assumptions</b> | The nominal mission supports seven crew members for 45 days with at least 60 days of critical provisions.                                                                                                                       |
| <b>Preliminary owner</b>    | Life Support Lead                                                                                                                                                                                                               |
| <b>Primary risk</b>         | A life-safety shortfall, contamination event, or insufficient margin may prevent crew authorization.                                                                                                                            |
| <b>Decision gate</b>        | Crewed Mission Readiness Review                                                                                                                                                                                                 |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                   |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.4.3 Crew Consumables and Waste Management

Parent element 1.4 Habitat and Crew Readiness

| WBS Dictionary Entry |                                                                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the crew consumables and waste management capability required to support the habitat and crew readiness outcome.                                                                                                      |
| Included work        | Food, potable water, hygiene, clothing, consumables, storage, inventory, waste processing, and contingency margins.                                                                                                             |
| Excluded work        | This branch owns crew-specific systems and readiness; common infrastructure serving multiple projects remains under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Food, water, hygiene, consumables, storage, and waste provisions support the approved duration.                                                                                                                                 |
| Key dependencies     | Verified site readiness; common utilities; transportation capability; approved crew concept and mission duration.                                                                                                               |
| Planning assumptions | The nominal mission supports seven crew members for 45 days with at least 60 days of critical provisions.                                                                                                                       |
| Preliminary owner    | Crew Logistics Lead                                                                                                                                                                                                             |
| Primary risk         | A life-safety shortfall, contamination event, or insufficient margin may prevent crew authorization.                                                                                                                            |
| Decision gate        | Crewed Mission Readiness Review                                                                                                                                                                                                 |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                   |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.4.4 Radiation Dust and Thermal Protection

Parent element 1.4 Habitat and Crew Readiness

| WBS Dictionary Entry |                                                                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the radiation dust and thermal protection capability required to support the habitat and crew readiness outcome.                                                                                                      |
| Included work        | Radiation shelter, shielding, exposure monitoring, dust exclusion and removal, insulation, and thermal protection.                                                                                                              |
| Excluded work        | This branch owns crew-specific systems and readiness; common infrastructure serving multiple projects remains under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Crew exposure and habitat contamination remain within approved radiation, dust, and thermal limits.                                                                                                                             |
| Key dependencies     | Verified site readiness; common utilities; transportation capability; approved crew concept and mission duration.                                                                                                               |
| Planning assumptions | The nominal mission supports seven crew members for 45 days with at least 60 days of critical provisions.                                                                                                                       |
| Preliminary owner    | Crew Protection Lead                                                                                                                                                                                                            |
| Primary risk         | A life-safety shortfall, contamination event, or insufficient margin may prevent crew authorization.                                                                                                                            |
| Decision gate        | Crewed Mission Readiness Review                                                                                                                                                                                                 |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                   |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.4.5 Crew Health Safety and Emergency Response

Parent element 1.4 Habitat and Crew Readiness

| WBS Dictionary Entry        |                                                                                                                                                                                                                                 |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the crew health safety and emergency response capability required to support the habitat and crew readiness outcome.                                                                                                  |
| <b>Included work</b>        | Medical capability, fire response, depressurization response, rescue, shelter-in-place, evacuation, and emergency drills.                                                                                                       |
| <b>Excluded work</b>        | This branch owns crew-specific systems and readiness; common infrastructure serving multiple projects remains under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Medical, fire, depressurization, shelter, rescue, and emergency procedures are verified.                                                                                                                                        |
| <b>Key dependencies</b>     | Verified site readiness; common utilities; transportation capability; approved crew concept and mission duration.                                                                                                               |
| <b>Planning assumptions</b> | The nominal mission supports seven crew members for 45 days with at least 60 days of critical provisions.                                                                                                                       |
| <b>Preliminary owner</b>    | Crew Safety and Medical Lead                                                                                                                                                                                                    |
| <b>Primary risk</b>         | A life-safety shortfall, contamination event, or insufficient margin may prevent crew authorization.                                                                                                                            |
| <b>Decision gate</b>        | Crewed Mission Readiness Review                                                                                                                                                                                                 |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                   |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.4.6 Crew Communications Operations and Safe Departure

Parent element 1.4 Habitat and Crew Readiness

| WBS Dictionary Entry |                                                                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the crew communications operations and safe departure capability required to support the habitat and crew readiness outcome.                                                                                          |
| Included work        | Crew voice and data, operating procedures, mission control interfaces, contingency return, ascent, and safe departure.                                                                                                          |
| Excluded work        | This branch owns crew-specific systems and readiness; common infrastructure serving multiple projects remains under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Crew communications, operating procedures, contingency return, and departure capability are ready.                                                                                                                              |
| Key dependencies     | Verified site readiness; common utilities; transportation capability; approved crew concept and mission duration.                                                                                                               |
| Planning assumptions | The nominal mission supports seven crew members for 45 days with at least 60 days of critical provisions.                                                                                                                       |
| Preliminary owner    | Crew Operations Lead                                                                                                                                                                                                            |
| Primary risk         | A life-safety shortfall, contamination event, or insufficient margin may prevent crew authorization.                                                                                                                            |
| Decision gate        | Crewed Mission Readiness Review                                                                                                                                                                                                 |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                   |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.4.7 Crew Mission Readiness and Training

Parent element 1.4 Habitat and Crew Readiness

| WBS Dictionary Entry |                                                                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the crew mission readiness and training capability required to support the habitat and crew readiness outcome.                                                                                                        |
| Included work        | Crew qualifications, simulations, maintenance skills, emergency drills, task certification, and readiness records.                                                                                                              |
| Excluded work        | This branch owns crew-specific systems and readiness; common infrastructure serving multiple projects remains under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Crew members demonstrate proficiency against approved mission and emergency scenarios.                                                                                                                                          |
| Key dependencies     | Verified site readiness; common utilities; transportation capability; approved crew concept and mission duration.                                                                                                               |
| Planning assumptions | The nominal mission supports seven crew members for 45 days with at least 60 days of critical provisions.                                                                                                                       |
| Preliminary owner    | Crew Readiness Lead                                                                                                                                                                                                             |
| Primary risk         | Evidence gaps or ambiguous criteria delay authorization at a decision gate.                                                                                                                                                     |
| Decision gate        | Crewed Mission Readiness Review                                                                                                                                                                                                 |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                   |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.5.1 Computing and Storage Architecture

Parent element 1.5 Lunar Data Center

| WBS Dictionary Entry        |                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the computing and storage architecture capability required to support the lunar data center outcome.                                                                                                                   |
| <b>Included work</b>        | Compute nodes, storage, workload allocation, redundancy, capacity, data integrity, expansion, and service architecture.                                                                                                          |
| <b>Excluded work</b>        | This branch owns data-center-specific equipment and controls; common utilities serving the broader site remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Compute, storage, workload, redundancy, and capacity architecture satisfy approved service requirements.                                                                                                                         |
| <b>Key dependencies</b>     | Approved workload requirements; site data; shared power and communications; thermal and maintenance interfaces.                                                                                                                  |
| <b>Planning assumptions</b> | Computing services may operate autonomously and in degraded modes within approved safety priorities.                                                                                                                             |
| <b>Preliminary owner</b>    | Data Center Architect                                                                                                                                                                                                            |
| <b>Primary risk</b>         | Requirement or interface changes propagate late rework across multiple projects.                                                                                                                                                 |
| <b>Decision gate</b>        | Data Center Commissioning Review                                                                                                                                                                                                 |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                    |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.5.2 Power Distribution and Energy Storage

Parent element 1.5 Lunar Data Center

| WBS Dictionary Entry        |                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the power distribution and energy storage capability required to support the lunar data center outcome.                                                                                                                |
| <b>Included work</b>        | Dedicated conversion, distribution, protection, switching, battery interfaces, power quality, metering, and load priorities.                                                                                                     |
| <b>Excluded work</b>        | This branch owns data-center-specific equipment and controls; common utilities serving the broader site remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Dedicated distribution, conversion, protection, and storage support required computing loads.                                                                                                                                    |
| <b>Key dependencies</b>     | Approved workload requirements; site data; shared power and communications; thermal and maintenance interfaces.                                                                                                                  |
| <b>Planning assumptions</b> | Computing services may operate autonomously and in degraded modes within approved safety priorities.                                                                                                                             |
| <b>Preliminary owner</b>    | Data Center Power Lead                                                                                                                                                                                                           |
| <b>Primary risk</b>         | Power, thermal, communications, cyber, or autonomous-control failures may reduce critical computing availability.                                                                                                                |
| <b>Decision gate</b>        | Data Center Commissioning Review                                                                                                                                                                                                 |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                    |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.5.3 Thermal Control and Heat Rejection

Parent element 1.5 Lunar Data Center

| WBS Dictionary Entry |                                                                                                                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the thermal control and heat rejection capability required to support the lunar data center outcome.                                                                                                                   |
| Included work        | Heat acquisition, transport, rejection, radiator sizing, controls, thermal monitoring, margins, and degraded modes.                                                                                                              |
| Excluded work        | This branch owns data-center-specific equipment and controls; common utilities serving the broader site remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Thermal transport and heat rejection maintain equipment within approved operating limits.                                                                                                                                        |
| Key dependencies     | Approved workload requirements; site data; shared power and communications; thermal and maintenance interfaces.                                                                                                                  |
| Planning assumptions | Computing services may operate autonomously and in degraded modes within approved safety priorities.                                                                                                                             |
| Preliminary owner    | Data Center Thermal Lead                                                                                                                                                                                                         |
| Primary risk         | Power, thermal, communications, cyber, or autonomous-control failures may reduce critical computing availability.                                                                                                                |
| Decision gate        | Data Center Commissioning Review                                                                                                                                                                                                 |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                    |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.5.4 Communications and Data Transfer

Parent element 1.5 Lunar Data Center

| WBS Dictionary Entry        |                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the communications and data transfer capability required to support the lunar data center outcome.                                                                                                                     |
| <b>Included work</b>        | Command, telemetry, data transfer, bandwidth allocation, latency, availability, protocols, and communications recovery.                                                                                                          |
| <b>Excluded work</b>        | This branch owns data-center-specific equipment and controls; common utilities serving the broader site remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Command, telemetry, data transfer, latency, and availability meet approved service criteria.                                                                                                                                     |
| <b>Key dependencies</b>     | Approved workload requirements; site data; shared power and communications; thermal and maintenance interfaces.                                                                                                                  |
| <b>Planning assumptions</b> | Computing services may operate autonomously and in degraded modes within approved safety priorities.                                                                                                                             |
| <b>Preliminary owner</b>    | Data Communications Lead                                                                                                                                                                                                         |
| <b>Primary risk</b>         | Undefined scope, interfaces, or acceptance evidence may delay completion.                                                                                                                                                        |
| <b>Decision gate</b>        | Data Center Commissioning Review                                                                                                                                                                                                 |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                    |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.5.5 Cybersecurity and Access Control

Parent element 1.5 Lunar Data Center

| WBS Dictionary Entry        |                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the cybersecurity and access control capability required to support the lunar data center outcome.                                                                                                                     |
| <b>Included work</b>        | Identity, access, segmentation, encryption, logging, monitoring, data protection, incident response, and recovery controls.                                                                                                      |
| <b>Excluded work</b>        | This branch owns data-center-specific equipment and controls; common utilities serving the broader site remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Identity, access, segmentation, monitoring, data protection, and incident response controls are accepted.                                                                                                                        |
| <b>Key dependencies</b>     | Approved workload requirements; site data; shared power and communications; thermal and maintenance interfaces.                                                                                                                  |
| <b>Planning assumptions</b> | Computing services may operate autonomously and in degraded modes within approved safety priorities.                                                                                                                             |
| <b>Preliminary owner</b>    | Cybersecurity Lead                                                                                                                                                                                                               |
| <b>Primary risk</b>         | Power, thermal, communications, cyber, or autonomous-control failures may reduce critical computing availability.                                                                                                                |
| <b>Decision gate</b>        | Data Center Commissioning Review                                                                                                                                                                                                 |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                    |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.5.6 Autonomous Monitoring and Fault Management

Parent element 1.5 Lunar Data Center

| WBS Dictionary Entry        |                                                                                                                                                                                                                                  |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the autonomous monitoring and fault management capability required to support the lunar data center outcome.                                                                                                           |
| <b>Included work</b>        | Health monitoring, anomaly detection, diagnosis, fault isolation, load shedding, autonomous safing, and automated recovery.                                                                                                      |
| <b>Excluded work</b>        | This branch owns data-center-specific equipment and controls; common utilities serving the broader site remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Monitoring, anomaly detection, isolation, load shedding, and automated recovery are verified.                                                                                                                                    |
| <b>Key dependencies</b>     | Approved workload requirements; site data; shared power and communications; thermal and maintenance interfaces.                                                                                                                  |
| <b>Planning assumptions</b> | Computing services may operate autonomously and in degraded modes within approved safety priorities.                                                                                                                             |
| <b>Preliminary owner</b>    | Autonomy and Reliability Lead                                                                                                                                                                                                    |
| <b>Primary risk</b>         | Power, thermal, communications, cyber, or autonomous-control failures may reduce critical computing availability.                                                                                                                |
| <b>Decision gate</b>        | Data Center Commissioning Review                                                                                                                                                                                                 |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                    |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.5.7 Commissioning Performance Validation and Recovery

Parent element 1.5 Lunar Data Center

| WBS Dictionary Entry |                                                                                                                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the commissioning performance validation and recovery capability required to support the lunar data center outcome.                                                                                                    |
| Included work        | Startup, workload tests, capacity tests, resilience scenarios, degraded operations, restoration, and acceptance evidence.                                                                                                        |
| Excluded work        | This branch owns data-center-specific equipment and controls; common utilities serving the broader site remain under 1.6. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | The data center passes performance, resilience, degraded-mode, and restoration demonstrations.                                                                                                                                   |
| Key dependencies     | Approved workload requirements; site data; shared power and communications; thermal and maintenance interfaces.                                                                                                                  |
| Planning assumptions | Computing services may operate autonomously and in degraded modes within approved safety priorities.                                                                                                                             |
| Preliminary owner    | Data Center Commissioning Lead                                                                                                                                                                                                   |
| Primary risk         | Undefined scope, interfaces, or acceptance evidence may delay completion.                                                                                                                                                        |
| Decision gate        | Data Center Commissioning Review                                                                                                                                                                                                 |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                    |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.6.1 Common Surface Power

Parent element 1.6 Shared Infrastructure and Resilience

| WBS Dictionary Entry |                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the common surface power capability required to support the shared infrastructure and resilience outcome.                                                                                                                |
| Included work        | Shared generation, distribution, switching, protection, storage interfaces, metering, allocation, and critical-load priorities.                                                                                                    |
| Excluded work        | Only services used by multiple projects belong here; project-dedicated equipment remains within its owning Level 2 element. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Shared generation, distribution, protection, storage interfaces, and priority rules are commissioned.                                                                                                                              |
| Key dependencies     | Allocated service demands from elements 1.3, 1.4, and 1.5; site architecture; priority and recovery rules.                                                                                                                         |
| Planning assumptions | Common services are shared only when doing so improves integration without creating unacceptable common-cause risk.                                                                                                                |
| Preliminary owner    | Shared Power Lead                                                                                                                                                                                                                  |
| Primary risk         | A shared-service or common-cause failure may affect multiple program elements simultaneously.                                                                                                                                      |
| Decision gate        | Integrated Infrastructure Readiness Review                                                                                                                                                                                         |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.6.2 Shared Communications Navigation and Timing

Parent element 1.6 Shared Infrastructure and Resilience

| WBS Dictionary Entry        |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the shared communications navigation and timing capability required to support the shared infrastructure and resilience outcome.                                                                                         |
| <b>Included work</b>        | Shared networks, relay services, positioning, navigation, timing, coverage, interoperability, and continuity.                                                                                                                      |
| <b>Excluded work</b>        | Only services used by multiple projects belong here; project-dedicated equipment remains within its owning Level 2 element. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Common communications, positioning, timing, and network services meet site requirements.                                                                                                                                           |
| <b>Key dependencies</b>     | Allocated service demands from elements 1.3, 1.4, and 1.5; site architecture; priority and recovery rules.                                                                                                                         |
| <b>Planning assumptions</b> | Common services are shared only when doing so improves integration without creating unacceptable common-cause risk.                                                                                                                |
| <b>Preliminary owner</b>    | Shared Network Lead                                                                                                                                                                                                                |
| <b>Primary risk</b>         | A shared-service or common-cause failure may affect multiple program elements simultaneously.                                                                                                                                      |
| <b>Decision gate</b>        | Integrated Infrastructure Readiness Review                                                                                                                                                                                         |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.6.3 Common Thermal Services

Parent element 1.6 Shared Infrastructure and Resilience

| WBS Dictionary Entry        |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the common thermal services capability required to support the shared infrastructure and resilience outcome.                                                                                                             |
| <b>Included work</b>        | Shared heat transport, rejection, allocation, interface control, metering, operating limits, and contingency capacity.                                                                                                             |
| <b>Excluded work</b>        | Only services used by multiple projects belong here; project-dedicated equipment remains within its owning Level 2 element. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Shared heat-transport or rejection services meet allocated loads and interface requirements.                                                                                                                                       |
| <b>Key dependencies</b>     | Allocated service demands from elements 1.3, 1.4, and 1.5; site architecture; priority and recovery rules.                                                                                                                         |
| <b>Planning assumptions</b> | Common services are shared only when doing so improves integration without creating unacceptable common-cause risk.                                                                                                                |
| <b>Preliminary owner</b>    | Shared Thermal Lead                                                                                                                                                                                                                |
| <b>Primary risk</b>         | A shared-service or common-cause failure may affect multiple program elements simultaneously.                                                                                                                                      |
| <b>Decision gate</b>        | Integrated Infrastructure Readiness Review                                                                                                                                                                                         |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.6.4 Logistics Spares and Maintenance Support

Parent element 1.6 Shared Infrastructure and Resilience

| WBS Dictionary Entry        |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the logistics spares and maintenance support capability required to support the shared infrastructure and resilience outcome.                                                                                            |
| <b>Included work</b>        | Spares, tools, inventory, maintenance data, material movement, storage, replenishment, repair, and cannibalization rules.                                                                                                          |
| <b>Excluded work</b>        | Only services used by multiple projects belong here; project-dedicated equipment remains within its owning Level 2 element. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Spares, tools, inventory, transport, maintenance data, and replenishment processes are ready.                                                                                                                                      |
| <b>Key dependencies</b>     | Allocated service demands from elements 1.3, 1.4, and 1.5; site architecture; priority and recovery rules.                                                                                                                         |
| <b>Planning assumptions</b> | Common services are shared only when doing so improves integration without creating unacceptable common-cause risk.                                                                                                                |
| <b>Preliminary owner</b>    | Surface Logistics Lead                                                                                                                                                                                                             |
| <b>Primary risk</b>         | A shared-service or common-cause failure may affect multiple program elements simultaneously.                                                                                                                                      |
| <b>Decision gate</b>        | Integrated Infrastructure Readiness Review                                                                                                                                                                                         |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.6.5 Dust Mitigation and Environmental Monitoring

Parent element 1.6 Shared Infrastructure and Resilience

| WBS Dictionary Entry        |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the dust mitigation and environmental monitoring capability required to support the shared infrastructure and resilience outcome.                                                                                        |
| <b>Included work</b>        | Environmental sensing, dust prevention, detection, removal, contamination control, trending, alerts, and response.                                                                                                                 |
| <b>Excluded work</b>        | Only services used by multiple projects belong here; project-dedicated equipment remains within its owning Level 2 element. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Site monitoring and dust prevention, detection, removal, and response capabilities are operational.                                                                                                                                |
| <b>Key dependencies</b>     | Allocated service demands from elements 1.3, 1.4, and 1.5; site architecture; priority and recovery rules.                                                                                                                         |
| <b>Planning assumptions</b> | Common services are shared only when doing so improves integration without creating unacceptable common-cause risk.                                                                                                                |
| <b>Preliminary owner</b>    | Environmental Operations Lead                                                                                                                                                                                                      |
| <b>Primary risk</b>         | A shared-service or common-cause failure may affect multiple program elements simultaneously.                                                                                                                                      |
| <b>Decision gate</b>        | Integrated Infrastructure Readiness Review                                                                                                                                                                                         |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.6.6 Emergency Power and Contingency Operations

Parent element 1.6 Shared Infrastructure and Resilience

| WBS Dictionary Entry        |                                                                                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope definition</b>     | Establish the emergency power and contingency operations capability required to support the shared infrastructure and resilience outcome.                                                                                          |
| <b>Included work</b>        | Emergency generation and storage, load priorities, safe states, black-start, contingency procedures, and restoration sequence.                                                                                                     |
| <b>Excluded work</b>        | Only services used by multiple projects belong here; project-dedicated equipment remains within its owning Level 2 element. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| <b>Acceptance criteria</b>  | Critical-load priorities, emergency power, safe states, and contingency procedures are verified.                                                                                                                                   |
| <b>Key dependencies</b>     | Allocated service demands from elements 1.3, 1.4, and 1.5; site architecture; priority and recovery rules.                                                                                                                         |
| <b>Planning assumptions</b> | Common services are shared only when doing so improves integration without creating unacceptable common-cause risk.                                                                                                                |
| <b>Preliminary owner</b>    | Contingency Operations Lead                                                                                                                                                                                                        |
| <b>Primary risk</b>         | A shared-service or common-cause failure may affect multiple program elements simultaneously.                                                                                                                                      |
| <b>Decision gate</b>        | Integrated Infrastructure Readiness Review                                                                                                                                                                                         |
| <b>Decomposition</b>        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

## 1.6.7 Integrated Resilience and Recovery Demonstration

Parent element 1.6 Shared Infrastructure and Resilience

| WBS Dictionary Entry |                                                                                                                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope definition     | Establish the integrated resilience and recovery demonstration capability required to support the shared infrastructure and resilience outcome.                                                                                    |
| Included work        | Cross-system failure scenarios, common-cause analysis, isolation, degraded operations, recovery, exercise evidence, and lessons learned.                                                                                           |
| Excluded work        | Only services used by multiple projects belong here; project-dedicated equipment remains within its owning Level 2 element. Detailed execution activities remain within approved lower-level work packages or schedule activities. |
| Acceptance criteria  | Cross-system disruption, isolation, degraded operation, and recovery scenarios are successfully demonstrated.                                                                                                                      |
| Key dependencies     | Allocated service demands from elements 1.3, 1.4, and 1.5; site architecture; priority and recovery rules.                                                                                                                         |
| Planning assumptions | Common services are shared only when doing so improves integration without creating unacceptable common-cause risk.                                                                                                                |
| Preliminary owner    | Program Resilience Lead                                                                                                                                                                                                            |
| Primary risk         | A shared-service or common-cause failure may affect multiple program elements simultaneously.                                                                                                                                      |
| Decision gate        | Integrated Infrastructure Readiness Review                                                                                                                                                                                         |
| Decomposition        | Decompose to Level 4 work packages before detailed scheduling, resource loading, and execution authorization.                                                                                                                      |

**Control note:** Ownership, dependencies, risks, and gate assignments are preliminary until approved through program baseline control.

# Review and Baseline Actions

*Required decisions before Level 4 planning*

## Review actions

- Assign each Level 3 entry to an accountable branch owner for validation.
- Confirm that inclusions and exclusions provide complete coverage without duplicated scope.
- Convert preliminary acceptance statements into measurable criteria linked to verification evidence.
- Validate cross-branch dependencies and capture them in the interface and integrated schedule baselines.
- Approve the recommended Level 4 decompositions and identify level-of-effort control elements.
- Assign named owners, budgets, resources, dates, and risk responses only after scope approval.

## Next authorized artifact

Develop Level 4 work-package definitions for the selected Level 3 elements, followed by the integrated master schedule, responsibility assignment matrix, and initial risk register. Each work package must have a single accountable owner, measurable completion criteria, estimated resources, and controlled interfaces.

**Revision record:** Version 0.1, 21 September 2026. Initial WBS Dictionary draft defining 40 Level 3 elements from WBS v0.2.